

Reproducibility protocol

Status: NOT READY - release identifiers and validated dependency lock pending

1. Scope and scientific lock

This protocol reconstructs the locked analysis environment and execution order without changing samples, donors, preprocessing, module definitions, models, multiple-testing families, random seeds, figures, conclusions, or evidence tiers. Raw files are read-only. The locked integrity result is `raw_data_modified = FALSE`.

2. Public inputs

Obtain GSE135251, GSE167523, GSE126848, GSE98793, GSE102556, GSE144136, GSE48350, GSE5281, and GSE33000 from GEO. Obtain the applicable Human Protein Atlas tissue-consensus release only if reproducing the descriptive HPA analysis. Record accession, source URL, retrieval date, platform, object type, and checksum. Do not commit third-party matrices.

3. Local staging

Create a local analysis root outside version control. Preserve the project-relative input structure expected by the locked scripts, including `01_raw_data`, `04_intermediate`, and verified donor-map inputs. Configure the root in `config.yaml` or the `CTMR_ANALYSIS_ROOT` environment variable. Store generated outputs under `results/generated`.

4. Environment

Use R 4.5.2 and the package versions recorded in `environment/package_versions.csv`. Generate a complete `renv.lock` in a clean project, restore it, and record `sessionInfo()`. The checked-in `renv.lock` is an audit placeholder and must be replaced before release.

5. Ordered workflow

Step	Folder	Primary output
1	01_data_download	Source files, inventory, hash verification
2	02_preprocessing	Dataset registry and normalization sensitivity
3	03_probe_mapping	Strict mapping and coverage sensitivity
4	04_module_scoring	Dataset-specific GSVA/ssGSEA scores
5	05_donor_models	Donor-aware and naive model tables
6	06_random_controls	B=2000 empirical negative controls
7	07_meta_analysis	Dataset effects, REML, Hartung-Knapp, CR2
8	08_module_preservation	500-permutation preservation results
9	09_HPA_analysis	Descriptive liver-expression attributes
10	10_figures_tables	Evidence matrix, figures, consistency audit

6. Verification gates

Confirm exact sample and donor counts, 24-module and 144-test BH families, separation of conventional REML and Hartung-Knapp results, B=2000 and seed 20260802, 500 preservation permutations, Tier A/B/C/D = 0/1/21/2, and absence of raw matrices or local paths. Recompute repository SHA-256 values after final identifiers are inserted.

7. Release gate

Release only after a clean restore test, verified GitHub URL, immutable tag, Zenodo DOI, final citation/metadata synchronization, and public/private scan. Until then the correct status is NOT READY.